Satellite Image Handling and Interpretation: A Study of Surat, Gujarat

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Introduction

Remote sensing is used to observe, analyze and monitor the Earth's surface using satellite-based measurements. Multispectral satellite imagery helps us to identify and interpret different land – cover features based on their spectral characteristics, spatial patterns and surface reflectance properties.

This study focuses on Surat, Gujarat and examines the application of multispectral satellite imagery for image pre – processing, spectral signature analysis and visual interpretation of land cover features. The analysis uses Landsat 9 Collection 2 Level – 1 and Level – 2 imagery, together with Sentinel – 2 Level – 2A imagery.

All the practical work was done using ArcGIS Pro. It involves acquisition and organization of satellite data, preparation of Area of Interest (AOI), radiometric processing, image resampling, spectral analysis and visual interpretation. A selected 1km X 1km window within Surat was digitized to identify and estimate the spatial distribution of different land cover classes.

Objectives of the Study

The main objectives of this study are:

1. Acquire and examine Landsat 9 and Sentinel 2 satellite imagery and understand their sensor characteristics.
2. To perform satellite image pre-processing, including reflectance conversion, band stacking, clipping and resampling.
3. To examine the spectral characteristics of selected land-cover classes using multispectral reflectance values.
4. To interpret natural colour, colour infrared and shortwave infrared composites.
5. To digitize and estimate the area distribution of selected land cover classes within a 1 km x 1km study window.
6. To make a thematic land cover map using satellite imagery and GIS based interpretation.

Data Acquisition and Sensor Profile

Satellite data acquisition

Three satellite data were acquired for the study of Surat, Gujarat: Landsat 9 Collection 2 Level-1, Landsat 9 Collection 2 Level-2 and Sentinel-2 Level-2A.

Landsat 9 datasets were obtained from the United States Geological Survey (USGS) EarthExplorer platform. The Level-1 product is the radiometrically calibrated and orthorectified image data. The Level -2 product is atmospherically corrected surface reflectance data. Both products are from same acquisition date, allowing the comparison of top-of-atmosphere and surface reflectance.

The Sentinel-2 Level 2A dataset was acquired form the Copernicus Data Space Ecosystem. It provides atmospherically corrected multispectral imagery with different spatial resolutions across its spectral bands.

The selected Landsat 9 scene was acquired on 7 May 2026, while the Sentinel-2A scene was acquired on 15 October 2026. Both datasets fall within the three-year acquisition period specified in the assignment.

Satellite and Sensor Characteristics

Parameter	Landsat 9 Level-1	Landsat 9 Level-2	Sentinel-2A Level-2A
Satellite	Landsat 9	Landsat 9	Sentinel-2A
Sensor	OLI-2 and TIRS-2	OLI-2 and TIRS-2	MSI
Acquisition date	7 May 2026	7 May 2026	15 October 2026
Product level	Collection 2 L1TP	Collection 2 L2SP	Level-2A
Scene / Product ID	`LC914804520261 27LGN00`	`LC09_L2SP_1480 45_20260507_20 260508_02_T1`	`S2A_MSIL2A_20261015T054241_N0513_ R005_T43QBD_20261015T103113`
Spatial resolution	30 m multispectral; 15 m panchromatic; 100 m thermal native	30 m surface reflectance	10 m, 20 m and 60 m, depending on band
Spectral resolution	11 bands	7 surface-reflectance bands	13 bands
Radiometric resolution	16-bit	16-bit stored data	12-bit
Temporal resolution	16 days	16 days	10 days for Sentinel-2A; 5 days for the two-satellite constellation

Selection of Satellite Sensors

Landsat 9 provides a consistent multispectral dataset with a 30m spatial resolution, making it suitable for examining relatively large and spatially distinct features such as airports.

Sentinel – 2 is particularly suitable for monitoring seasonal changes over a year because of its higher revisit frequency.

Pre – processing

Area of interest

A 60.85 km² area of interest (AOI) for Surat, India was delineated to provide a defined study area for spatial analysis.

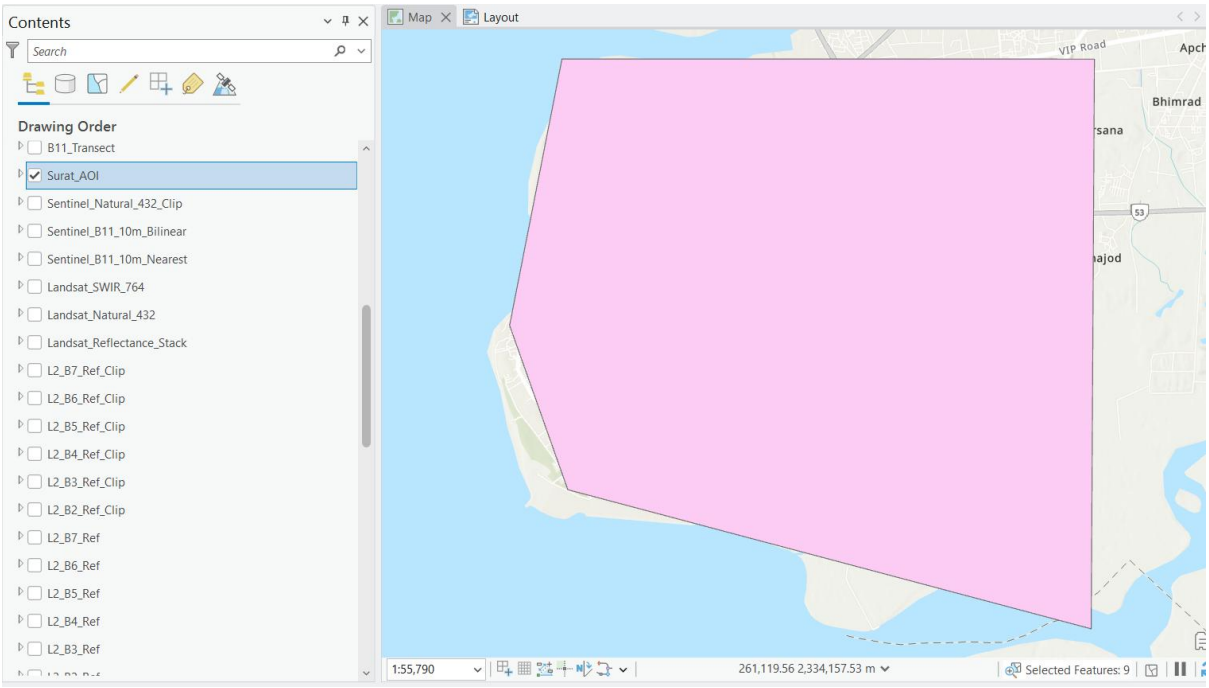


Figure 1 Area of Interest (AOI) for Surat

The coordinate reference system selected for the analysis was **WGS 1984 UTM Zone 43N (EPSG:32643).** A projected coordinate system was used to maintain measurements in meters and support accurate area calculations, spatial analysis and image alignment.

The details of AIO and Coordinate reference system is given below:

Parameter	**Details**
Study area	Surat, Gujarat, India
AOI name	`Surat AOI`
AOI area	60.847346 km²
Projection	Universal Transverse Mercator (UTM)
Datum	WGS 1984
UTM zone	43N
EPSG code	32643
Linear unit	Metre

All the satellite datasets were checked for spatial reference consistency. The Sentinel – 2 imagery was already available in WGS 1984 UTM Zone 43N (EPSG:32643), matching the Landsat 9 data and the study – area coordinate system.

TOA Reflectance Calculation and Comparison with Surface Reflectance

The Level – 1 digital number (DN) values were converted to Top of atmosphere (TOA) reflectance using the reflectance rescaling coefficients provided in Landsat 9 metadata file (MTL).

The calculation involves two steps:

1. Apply the reflectance rescaling coefficients

$$\rho'_\lambda = M_\rho \times Q_{cal} + A_\rho$$

2. Correct for solar elevation angle

$$\rho_\lambda = \frac{\rho'_\lambda}{\sin(\theta_{SE})}$$

Where,

- ρ'_λ = TOA reflectance corrected for solar elevation
- M_ρ = Reflectance multiplicative scaling factor
- A_ρ = Reflectance additive scaling factor
- Q_{cal} = Level-1 pixel DN value
- θ_{SE} = Solar elevation angle

For Landsat 9 scene acquired on 7 May 2026, the metadata specifies a multiplicative factor of 0.00002, an additive factor of – 0.1, and a solar elevation angle of 67.85125473° for the blue and red bands.

Comparison with Level-2 Surface Reflectance

The calculated TOA reflectance values compared with the corresponding Level-2 Surface Reflectance values for the same land- cover patches.

Sample patch	Land-cover type	Band	Level-1 DN	TOA reflectance	Level-2 SR
1	Ocean water	Blue (B2)	10,965	0.128762	0.020083
1	Ocean water	Red (B4)	8,814	0.081861	0.04233
2	Dense vegetation	Blue (B2)	11,353	0.145237	0.063395
2	Dense vegetation	Red (B4)	10,182	0.122197	0.1003
3	Built-up	Blue (B2)	16,529	0.229322	0.169298
3	Built-up	Red (B4)	17,977	0.272185	0.275172

Three represented sample location were selected from Landsat 9 imagery to represent ocean water, dense vegetation and built-up land. Sample points were created for these land cover categories and corresponding pixel values were examined using the Explore tool in ArcGIS Pro.

For each sample, the Digital Number (DN) values for the blue (B2) and red (B4) bands were obtained from Landsat 9 Level 1 imagery. TOA reflectance was calculated using multiplicative and additive reflectance scaling coefficients given in Level 1 MTL metadata file.

The calculated TOA reflectance values were compared with the corresponding Level 2 Surface Reflectance values for the blue and red bands across the three selected land cover categories.

TOA reflectance represents the radiation measured at the top of atmosphere and is affected by atmospheric scattering and absorption. Level 2 Surface Reflectance does atmospheric correction to reduce these effects and provide a closer representation of actual surface reflectance. Consequently, differences between the two values vary across spectral bands and land – cover types.

Conversion of Level-2 Bands to Surface Reflectance

The Landsat 9 Collection 2 Level – 2 imagery was used to make surface reflectance. Bands B2 to B7 were selected for processing. The original Digital Number values were converted to surface reflectance using the scaling coefficients provided in Level 2 MTL metadata file.

The conversion formula used was: (DN × 0.0000275 − 0.2, fill pixels as NoData)

Each Band from B2 to B7 was converted using Raster Calculator. The resulting rasters were then clipped to the defined study area boundary (Surat_AOI).

The six processed reflectance bands were combined using Composite Bands tool. The bands were stacked in order from B2 to B7, producing a multispectral reflectance composite for spectral analysis and visual interpretation.

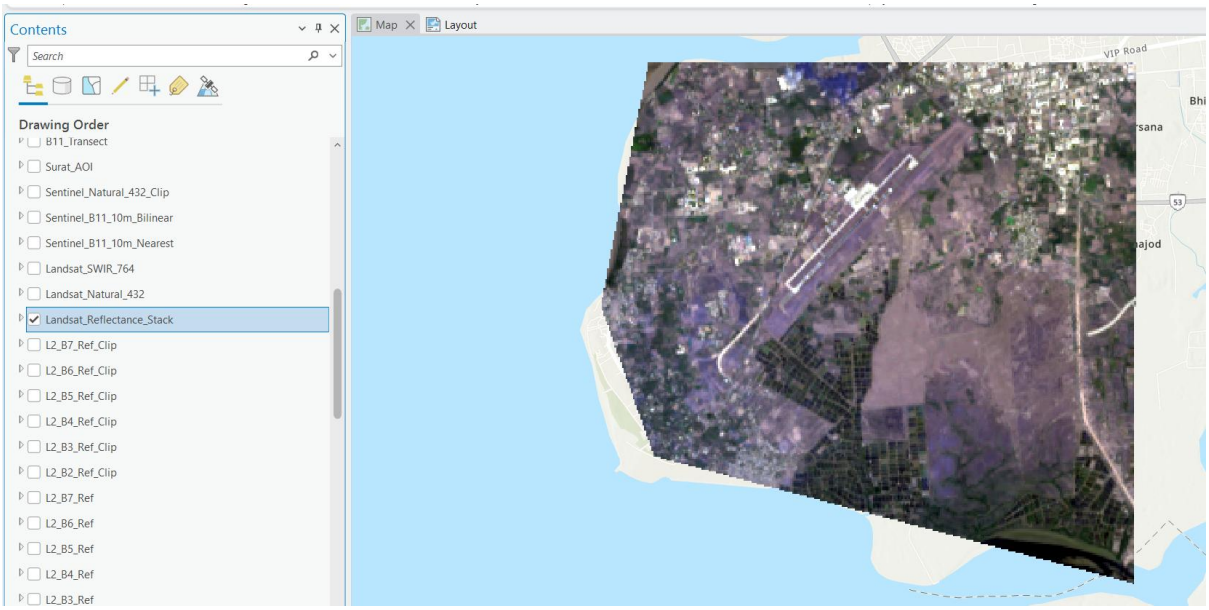


Figure 2 Landsat Level-2 Reflectance Stack

Sentinel-2 Band Resampling

Sentinel-2 Band 11 has a spatial resolution of 20m, whereas visible and near infrared bands have a spatial resolution of 10m. I used two resampling techniques: Nearest Neighbour and Bilinear Interpolation for resampling. Nearest Neighbour assigns the value of the closest pixel to the output pixel, while Bilinear Interpolation estimates the output using a weighted average of the four nearest input pixels.

The two resample outputs were compared at an image edge to examine differences in pixel value and the appearance of the boundary.

The transect – based visual comparison shows difference between two resampling techniques. Nearest Neighbour retains the original pixel values, resulting in more distinct and abrupt pixel transitions along the image edge. Bilinear Interpolation calculated the weighted averages from neighbouring pixels, thereby producing a smoother transition but changing the original pixel values. Therefore, we can say that Nearest Neighbour preserves the original spectral values, while Bilinear Interpolation provides a smoother spatial representation.

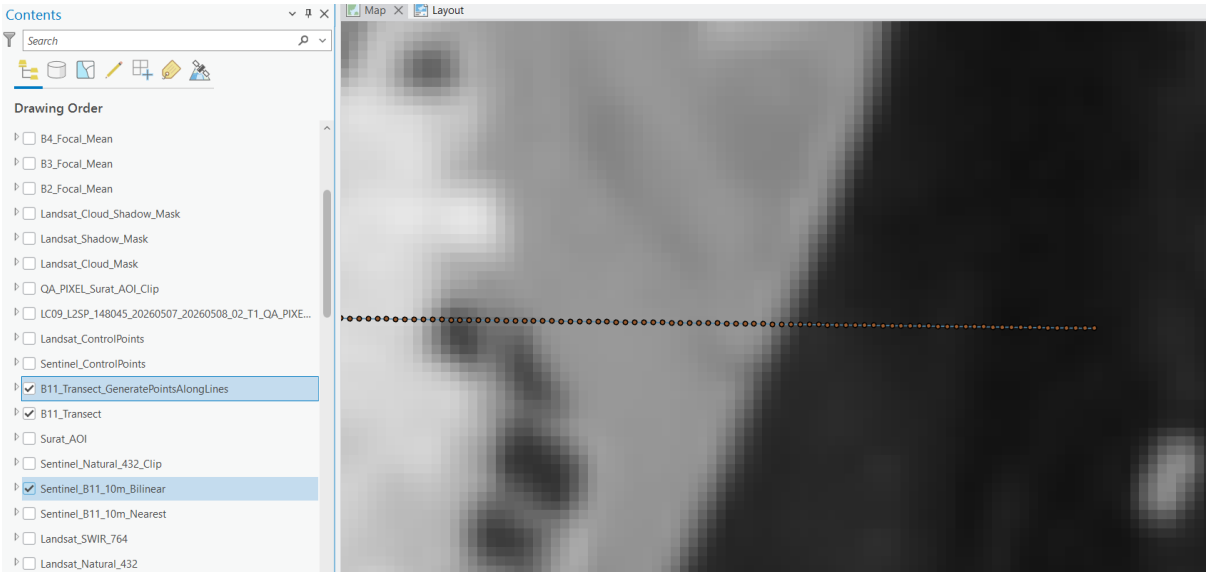


Figure 3 Sentinel B11_10m_Bilinear

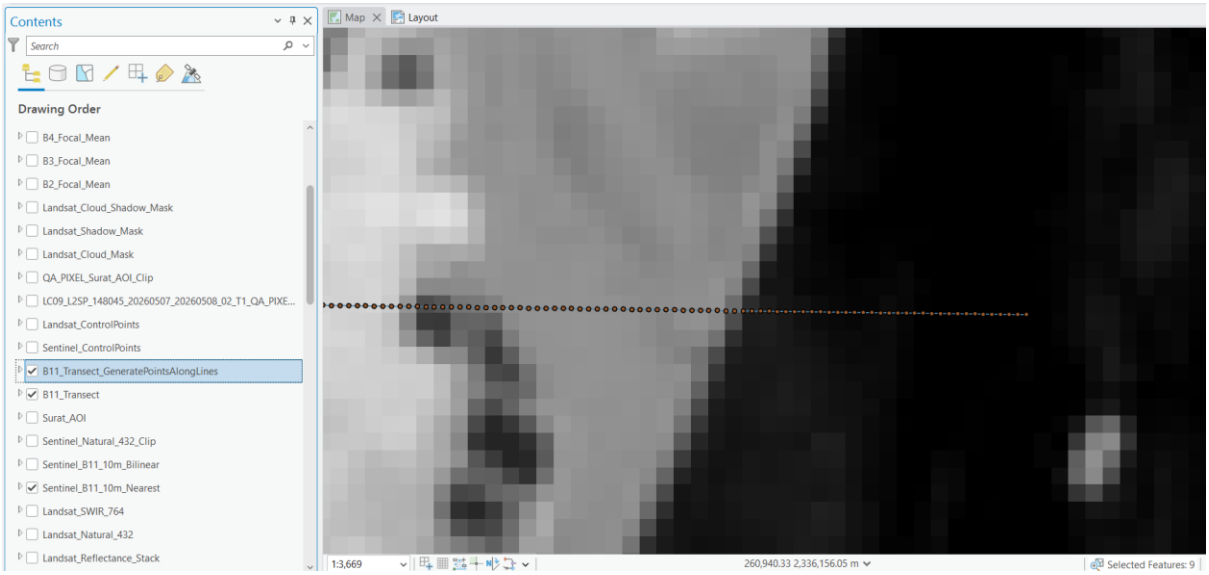
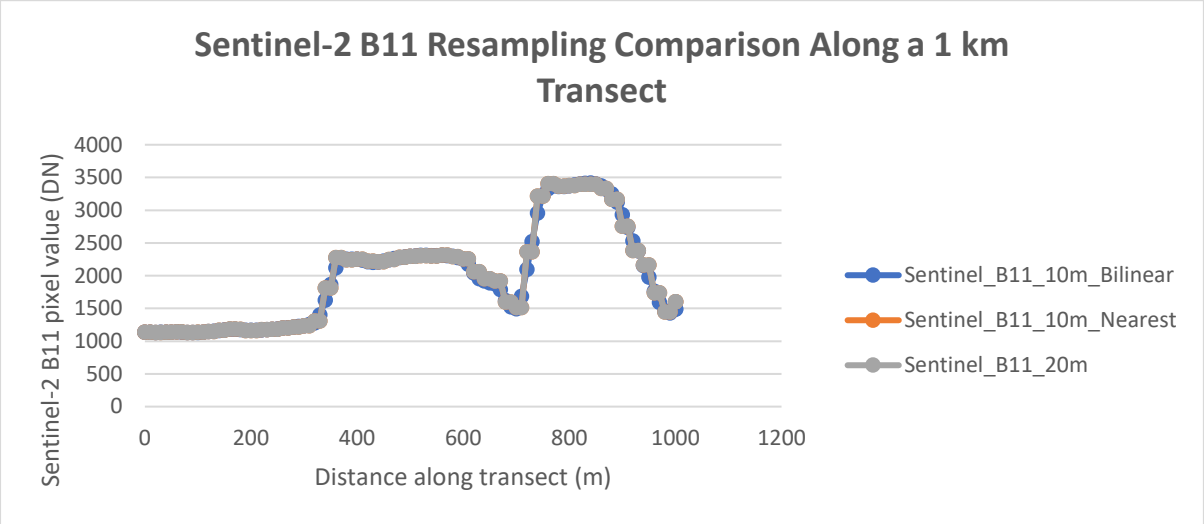


Figure 4 Sentinel B11_10m_Nearest

Example from actual data:



Point	Nearest Neighbour	Bilinear	Absolute difference
1	1,133	1,136	3
2	1,133	1,134	1
3	1,130	1,132	2
35	1,811	1,623	188
73	2,362	2,094	268
75	3,213	2,952	261
96	2,161	1,974	187

The result showed that 101 out of 102 sample points had different pixel values between two methods. The **mean absolute difference was 35.58**, while the **maximum absolute difference was 268**. The correlation between two sets of values was approximately 0.996, indicating that both methods retained a broader similar spatial pattern despite differences in individual pixel values.

Landsat – Sentinel offset and Cloud/Shadow:

Control point	ΔX (m)	ΔY (m)	Offset (m)
CP01 – Runway-taxiway junction	-3.7	2.17	4.29

CP02 – Major road intersection	-8.27	-2.76	8.72
CP03 – Curved road junction	-8	3.42	8.7

Mean positional offset: **approximately 7.24m**

RMSE: **approximately 7.53m**

The positional offset is calculated as:

$$d = \sqrt{(\Delta X)^2 + (\Delta Y)^2}$$

Cloud and cloud shadow assessment:

From the data it is reported

Assessment parameter	Result
Total AOI pixels assessed	67,533
Cloud-flagged pixels	0
Shadow-flagged pixels	0
Combined cloud/shadow pixels	0
Cloud and shadow coverage	0%

The percentage of cloud cover is calculated as:

$$Cloud\ coverage\ = \frac{Cloud\ and\ shadow\ pixels}{Total\ AOI\ pixels} \times 100$$

Since no cloud or shadow pixels were flagged in the assessed AOI, the reported coverage is 0%.

Spectral signature

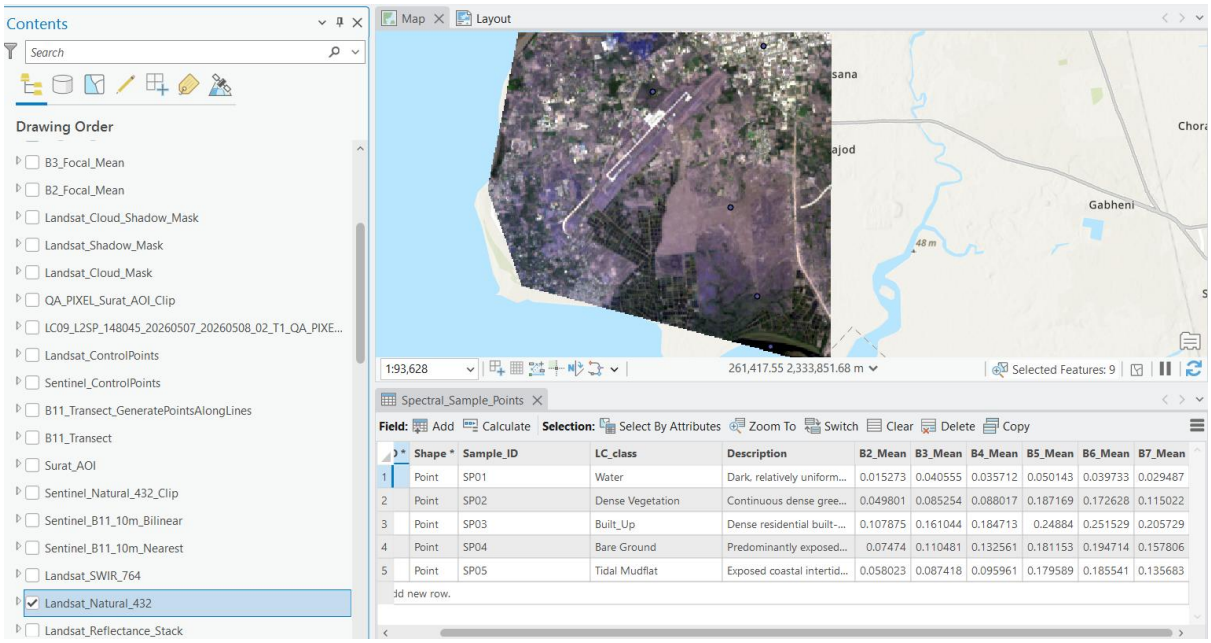
Spectral signature analysis was performed to see the variation in surface reflectance across different land cover categories using Landsat 9 Level 2 imagery. Five representative land cover features were selected: Water, Dense Vegetation, Built up, Bare Ground and Tidal Mudflat.

Representative sample locations were identified for each of five land cover categories. A 3 × 3-pixel neighborhood was used in order to obtain the mean reflectance values for each class across the six selected spectral bands (B2 – B7).

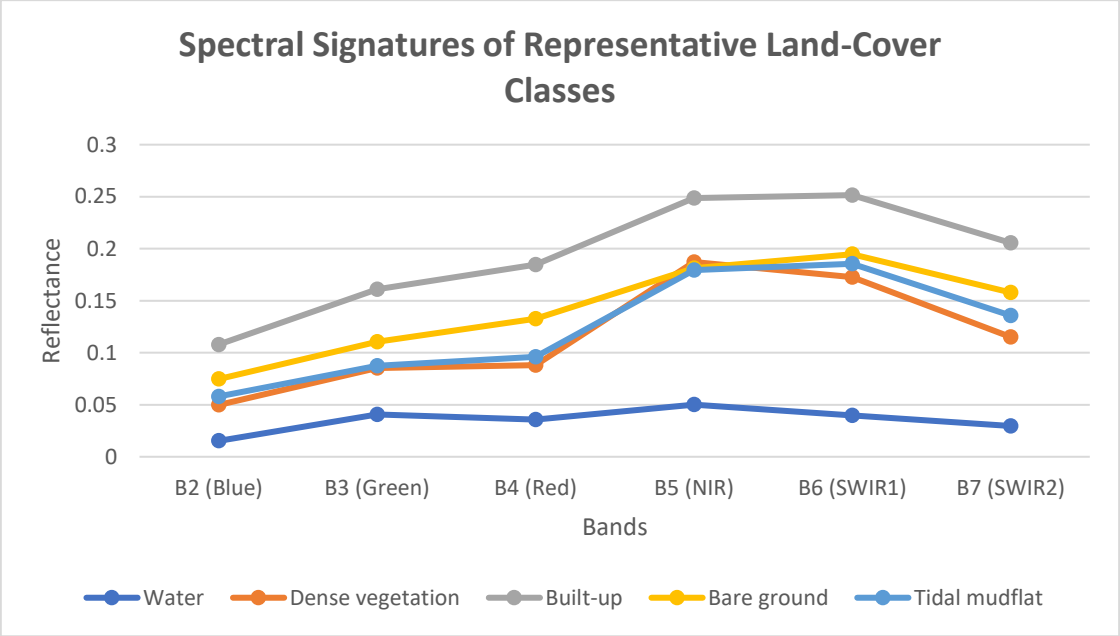
The focal mean operation was performed in ArcGIS in order to calculate the average reflectance within each 3×3-pixel neighborhood. This gave a representative spectral value for each sampled land cover category while reducing the influence of individual pixel level variation.

The six bands used in the analysis were:

- B2 – Blue
- B3 – Green
- B4 – Red
- B5 – Near Infrared (NIR)
- B6 – Shortwave Infrared 1 (SWIR 1)
- B7 – Shortwave Infrared 2 (SWIR 2)



Land-cover class	B2 (Blue)	B3 (Green)	B4 (Red)	B5 (NIR)	B6 (SWIR1)	B7 (SWIR2)
Water	0.0153	0.0406	0.0357	0.0501	0.0397	0.0295
Dense vegetation	0.0498	0.0853	0.088	0.1872	0.1726	0.115
Built-up	0.1079	0.161	0.1847	0.2488	0.2515	0.2057
Bare ground	0.0747	0.1105	0.1326	0.1812	0.1947	0.1578
Tidal mudflat	0.058	0.0874	0.096	0.1796	0.1855	0.1357



Vegetation response: Dense vegetation shows a noticeable increase in reflectance band (B4: 0.0880) to the near infrared band (B5: 0.1872). This characteristic increase is associated with strong infrared reflection by healthy vegetation. Reflectance decreases in the SWIR bands, with values of 0.1726 in B6 and 0.1150 in B7.

Water response: Water has lowest reflectance among the five sampled classes across all the bands. Its reflectance is low in NIR and SWIR regions, which help to distinguish water from terrestrial land cover categories.

Similarity between classes: Dense vegetation and tidal mudflat display relatively similar reflectance patterns, particularly in the blue, green, red and NIR bands. Their differences become more noticeable in the SWIR bands. This similarity may make their classification more challenging using individual spectral bands.

Best sampling band: B6 (SWIR1) gives the widest reflectance range among the five sampled classes, from 0.0397 for water to 0.2515 for built up. It therefore offers strong overall separation between the sampled classes.

Visual Interpretation

Three multispectral composites were prepared using Landsat 9 Level 2 surface reflectance bands.

1. **Natural Colour Composite (4-3-2):** Red, Green and Blue bands were assigned to the corresponding display channels to produce a natural-colour representation of the study area.

2. **Colour Infrared Composite (5-4-3):** Near-infrared, red and green bands were used to highlight vegetation and distinguish it from other surface features.
3. **SWIR Composite (7-6-4):** Shortwave Infrared 2, Shortwave Infrared 1 and red bands were combined to improve visual differentiation between vegetation, built-up surfaces and bare land.

Land-cover class	Natural Colour (4-3-2)	Colour Infrared (5-4-3)	SWIR (7-6-4)
Water	Generally dark	Dark	Dark
Dense vegetation	Green	Bright red	Variable, depending on moisture
Built-up	Grey or light-toned	Cyan, grey or blue	Often distinguishable through contrasting tones
Bare ground	Brown or light-toned	Pale or cyan-toned	Often bright
Tidal mudflat	Brown or grey	Variable, often muted	Distinguishable through moisture-related tonal differences

Land cover digitalisation

Land-cover class	Number of polygons	Area (ha)
Agricultural land	1	8.097
Barren land	1	2.970
Built-up	3	19.383
Dense vegetation	3	19.733
Fallow land	1	2.203
Total digitised	9	52.386

A 1 km × 1 km interpretation window, covering 100 hectares, was selected within the Surat study area. The window was digitised in ArcGIS using the Colour Infrared (5-4-3) composite as the primary visual reference.

Land-cover features were manually delineated into polygons based on their visual characteristics and assigned corresponding land-cover categories. The digitised classes included agricultural land, barren land, built-up areas, dense vegetation and fallow land.

Four representative polygons were selected from the digitised land-cover layer for detailed visual interpretation. Their classification was assessed using six elements of visual interpretation: tone, texture, pattern, shape, shadow and association.

The Colour Infrared (5-4-3) composite was used as the primary reference for interpreting the selected polygons. These elements were considered together to distinguish land-cover features and assess the confidence of the assigned classifications.

The two low confidence polygons identified from attribute tables were: **Land cover class** and **Confidence**.

Four representative polygons were selected from the land-cover classification layer for detailed interpretation. The polygons represent agricultural land, barren land, built-up land and dense vegetation. Their identification was based on the combined assessment of tone, texture, pattern, shape, shadow and association in the Colour Infrared composite.

Interpretation element	Polygon 9 – Agricultural land	Polygon 11 – Barren land	Polygon 6 – Built-up	Polygon 15 – Dense vegetation
Tone	Variable red and brown tones in CIR	Muted or relatively low vegetation response in CIR.	Mixed cyan, grey and other tones depending on surface materials.	Bright red or pink tones, indicating vegetation response.
Texture	Relatively variable texture associated with cultivated fields.	Comparatively uniform or coarse exposed-surface texture.	Heterogeneous texture due to buildings and other urban features.	Relatively continuous vegetation texture.
Pattern	Field-like arrangement, potentially associated with agricultural plots.	Irregular distribution of exposed ground.	Clustered and irregular arrangement of urban structures.	Patches of relatively continuous vegetation cover.
Shape	Irregular field boundaries.	Irregular polygon boundary.	Geometric and irregular built-up forms.	Irregular vegetation patch.
Shadow	Limited distinct shadow evidence at the available image scale.	No prominent shadow evidence identified.	Shadows may occur around buildings and other structures.	Limited distinct shadow evidence at the available image scale.
Association	Associated with surrounding agricultural fields and open land.	Associated with exposed ground and adjoining land-cover types.	Associated with roads, settlements and other urban features.	Associated with surrounding vegetation patches and open land.
Classification confidence	Low	Low	High	High
Area (ha)	8.097	2.970	5.886	3.016

Limitation in spatial coverage:

Although the selected interpretation window covered 100 hectares, the digitisation focused on delineating representative land-cover patches that could be visually distinguished with reasonable confidence in the Landsat Colour Infrared composite. Consequently, only 52.386 hectares were covered by the nine digitised polygons, leaving the remaining 47.614 hectares unclassified.

Map

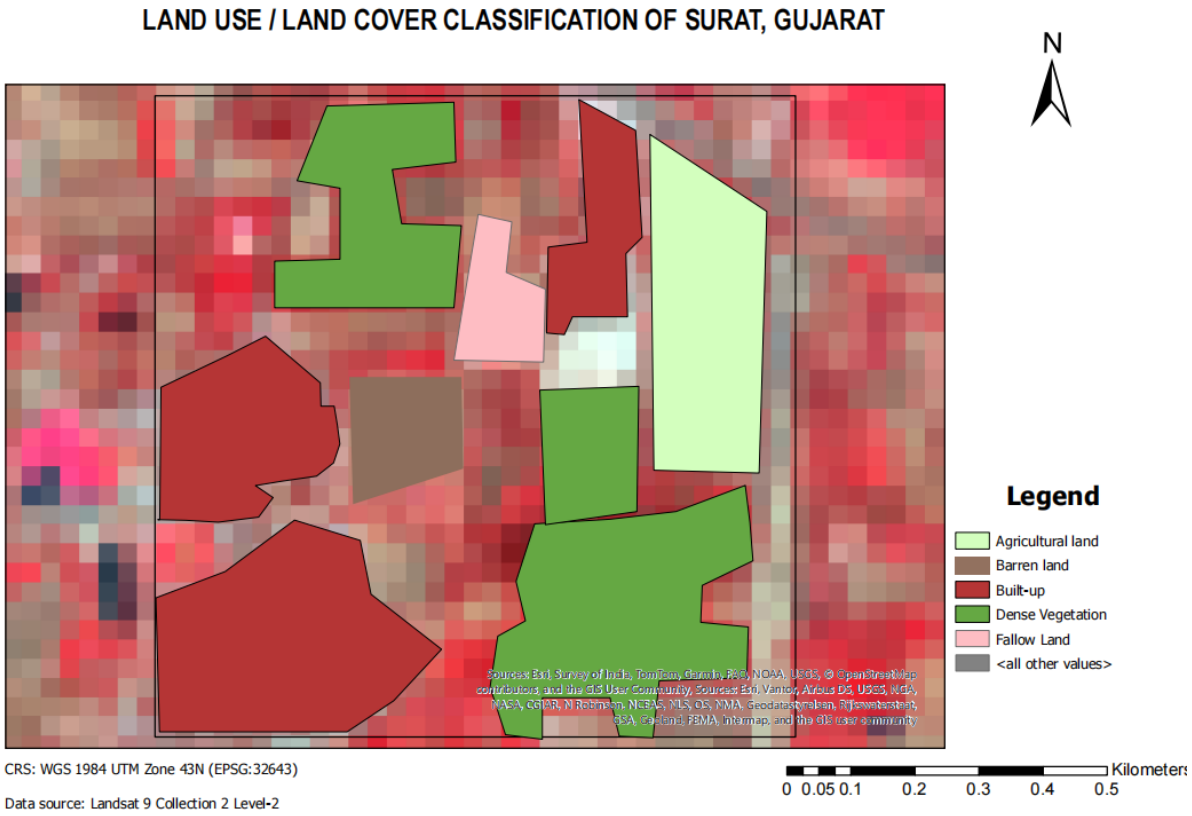


Figure 5 Land use/ Land cover classification of selected 1km x 1km window in Surat, Gujarat

Drive link for ArcGIS work:

https://drive.google.com/file/d/1PtTdSg-gX2N3Jvjakwk-RxlRjI-VV3HF/view?usp=sharing